Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/27/2006
PI Name:	Zhao, Yongliang Ph.D.		
Project Title:	Functional Role Of The Betaig-H3 Gene In High-Energy Heavy Ions-Induced Carcinogenesis		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shao, Genze (Columbia University) Berenguer, Jessica (Columbia University)		
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	Understanding the cellular and molecular mechanism of radiation carcinogenesis induced by high-energy heavy ions is important for both space radiation protection as well as design of interventional approaches. Our previous data demonstrate that loss of expression of Betaig-h3 gene is causally linked to tumorigenic process in high-LET, alpha particle-irradiated human papillomavirus-immortalized human bronchial epithelial (BEP2D) cells. Furthermore, loss of Betaig-h3 expression is common in many human tumor cell lines and reintroduction of this gene into human lung tumor cells completely abrogates their tumorigenicity, suggesting that Betaig-h3 gene may function as a tumor suppressor in a variety of human tumors. The newly established telomerase (hTERT)-immortalized human bronchial epithelial (hTERT-SAEC) cell model that has normal p53 and p16 functions represents an attractive alternative to further ascertain the mechanism involved in radiation carcinogenesis. In this study, we propose to examine the functional role of Betaig-h3 gene in high-energy heavy ions-induced carcinogenesis using the hTERT-SAEC cells and Betaig-h3		

Task Description:	knock-out mice. The proposal has 3 main aims: I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. Using the Betaig-h3 null mice model, we will address the following questions: (1) whether mice with disruption of this gene predispose to cancer; (2) Is the incidence of transformation induced by heavy ions higher in Betaig-h3 null embryo fibroblast cells than in wild type cells? (3) Does loss of Betaig-h3 gene in embryo fibroblast cells affect radiation sensitivity and apoptosis induced by high-energy heavy ions? I) To establish the tumorigenic cell model by irradiating hTERT-SAEC cells using beams of charged heavy ions; II) To demonstrate that loss of Betaig-h3 expression is causally related to the malignant transformation process; and III) To generate mouse embryonic stem cells and knock-out mice with disrupted Betaig-h3 gene. The data generated from the proposed studies will provide important insights into the role of Betaig-h3 gene in heavy ion-indued carcinogenesis
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cancer induction by space radiation is a major concern for manned space exploration. Among the various particulate components of ionizing radiation in space, high energy (HZE) heavy ions have been of special concern to radiobiologists and NASA administrators mainly because they contribute about 1% of the particle radiation field in frequency whereas they produce a considerable amount of the whole dose-equivalent because of their large energy deposition and high LET. For the acute effects like cell inactivation are of not so direct interest, late effects including genetic changes and cancer formation are more important for risk estimation. However, for this biological end-point there is no direct human evidence available. The research from this funding project will provide the important information ragarding the malignant transformation and molecular alterations induced by heavy ions.
Task Progress:	1. Human hTERT-immortalized SAEC cells has been irradiated by a single dose of 60 cGy of 56Fe ions. Following 5 months of continuous culture, tumorigenic potential was evaluated in immuno-suppressed nude mice. 4/5 of mice that was inoculated subcutaneously with the irradiated cells formed progressively growing tumors at four months, whereas no tumors were found in the mice injected with parental SAEC-hTERT cells. Two independent tumor cell lines have been established from the tumor nodules. 3. A good correlation has been observed between promoter hyper methylation and lost expression of Betaig-h3 gene in human tumor cell lines and primary lung tumors. 3. Homozygous Betaig-h3 targeting mice were born in normal mendelian frequencies and showed no differences in size, behavior or fertility from wild-type mice, however, had a increased frequency of spontaneous tumor growth. In contrast, no tumor was found in wild type control animals.
Bibliography Type:	Description: (Last Updated: 09/29/2008)
Abstracts for Journals and Proceedings	Shao, G; Piao, C.Q.; Hei, T.K.; Zhao, Y.L. "Malignant transformation of hTERT-immortalized human small airway epithelial cells induced by high LET heavy ions. " The 4-th International Workshop on Space Radiation Research and 17th Annual NASA Space Radiation Health Investigators' Workshop. Moscow-St. Petersberg, Russia. , Jun-2006
Articles in Peer-reviewed Journals	Zhao, Y.L.; Shao, G.; Piao, C.Q.; Berenguer, J and Hei, T.K. "Downregulation of Betaig-h3 gene is involved in the tumorigenesis in human bronchial epithelial cells induced by heavy-ion radiation" Radiation Research, 162, 655-659 , Dec-2004
Articles in Peer-reviewed Journals	Zhao, Y.L.; El-Gabry, M.; Hei, T.K. "Loss of Betaig-h3 protein is frequent in primary lung carcinoma and related to tumorigenic phenotype in lung cancer cells. " Molecular Carcinogenesis, 45:84-92 , Feb-2006
Articles in Peer-reviewed Journals	Shao, G.; Berenguer, J.; Borczuk, A.C.; Powell, C.A.; Hei, T.K.; Zhao, Y.L. "Epigenetic inactivation of Betaig-h3 gene in human cancer cells" Cancer Research, 66(9) , May-2006
Presentation	Shao, G.; Hei, T.K. and Zhao, Y.L. "Transformation of hTERT-immortalized human bronchial epithelial cells induced by high LET heavy ions" 16th annual Space Radiation Health Investigators' workshop. May 15-18, Port Jefferson, New York. May-2005